

Aditya Sreenath

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[LinkedIn](#) | [Github](#) | [Portfolio](#)

SUMMARY

Data Scientist & Engineer with 5+ years of experience in production data automation. Holding an **MS in Data Science** from the University of Michigan, specializing in **predictive modeling, time-series forecasting, and statistical analysis**. Proven track record of architecting high-volume **ETL pipelines** for enterprise clients, now focused on deploying advanced analytical models to drive business intelligence.

EDUCATION

University of Michigan - Dearborn | GPA: 3.81/4.0

Dearborn, MI, USA

Master of Science in Data Science

Sep 2024 – Apr 2026

- **Relevant Graduate Coursework:** Probability and Statistical Model, Applied Regression Analysis, Deep Learning, Introduction to Big Data, Big Data Analysis and Visualization, Database Systems, Cloud Computing, Project Management

SRM Institute of Science and Technology | CGPA: 7.60/10.00

Chennai, TN, India

Bachelor of Technology in Software Engineering

June 2015 – May 2019

PROFESSIONAL EXPERIENCE

Automation & Data Engineer | Capgemini (Client: Synchrony Financial)

June 2019 – Aug 2024

- Engineered a stateful automation system processing over 1,000 daily marketing campaign emails, delivering an estimated **\$1M in annual ROI** by maintaining dataset synchronization and eliminating manual errors.
- Developed robust **ETL workflows** to extract and validate data from diverse sources up to **5GB+**, implementing advanced exception-handling to manage complex file formats and maintain data integrity.
- Architected an end-to-end validation system cross-referencing JIRA data against QA standards, reducing validation time from **2 hours to 12 minutes** per release and enabling a 4-person team to handle **surges of up to 50 concurrent releases** during peak delivery windows.
- Led the migration of over 10 legacy automation bots to a cloud-ready **A360 architecture**, maintaining **100% service availability** throughout the transition.
- Provided technical mentorship to new joiners on **engineering best practices** and peer-review protocols, accelerating onboarding and improving the overall code quality of the automation pipeline.

SKILLS

Languages & Analytics Tools: Python (Pandas, PySpark, NumPy, Scikit-learn), SQL, Snowflake, R, Power BI, Git, Automation Anywhere (A360).

Data Engineering & Cloud (AWS): Kinesis Video Streams, Rekognition, Lambda, DynamoDB, API Gateway, S3, Apache Spark, Hadoop, Database Design & Normalization, ETL Pipelines.

Machine Learning & Statistical Methods: Predictive Modeling, Time-Series Forecasting (SARIMAX, LSTM), Random Forest, Logistic Regression, PyTorch, Hypothesis Testing (T-tests/ANOVA/Chi-squared), Differential Privacy.

PROJECT EXPERIENCE

Detroit Crime Analysis & Predictive Policing

- Processed over 750,000 historical records using **Pandas**, **Geopandas**, and **DBSCAN** clustering to identify high-density crime hotspots and develop interactive spatial-temporal dashboards.
- Conducted a comparative analysis of **SARIMAX** and **LSTM** architectures to model complex temporal patterns, delivering actionable trend insights to replace legacy, reactive forecasting methods.

Olist E-Commerce Analytics Platform

- Built an end-to-end analytics pipeline ingestive of multi-table relational schemata, utilizing **Python** and **SQL** to clean, merge, and transform raw transactional data into an optimized Kimball star schema.
- Designed an interactive **PowerBI** dashboard tracking critical commercial metrics, identifying key logistics bottlenecks to improve delivery success rates.

Healthcare Data Privacy & Governance

- Engineered a Differential Privacy framework using Laplace and Gaussian mechanisms to secure 300k+ patient records against re-identification attacks while maintaining 99%+ analytical utility.
- Utilized **Root Mean Square Error (RMSE)** to quantify the privacy-utility trade-off, ensuring that noise levels (0.016) effectively masked sensitive attributes like admission deposits and BMI without degrading model performance.

Smart Door Authentication System

- Architected a real-time facial recognition system using **AWS Kinesis Video Streams** and **Rekognition**, integrating **Lambda functions** and API Gateway to automate video ingestion and owner authorization workflows.
- Integrated **DynamoDB** to store authorized user profiles and managed **SNS/SQS** notification protocols to trigger instantaneous alerts for unrecognized subjects.

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- Automation Anywhere Masters Certification